



Instructor: Date:

Learner: Regional Branch:

Program: Center:

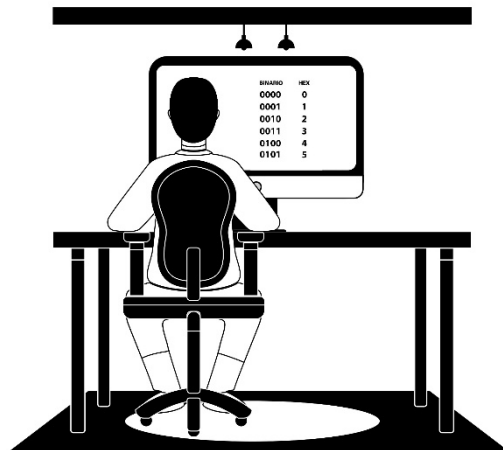
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Let's Start!

Become an expert in programming.



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can identify the concept of computer programming.
- I can understand the concept of computer programming.
- I can exchange information about computer programming.

Ask your partner:

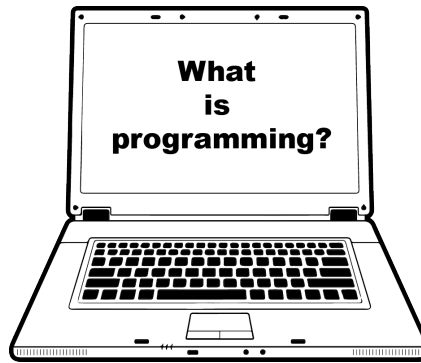
- In your opinion, what is the usefulness of programming systems?
- Do you know how to design the architecture of a program? Explain.





Let's Explore!

Ana and Carlos talk about programming.



Instructions: Click on the play button to listen to the audio.

Carlos: Good afternoon, Ana.

Ana: Hi, Carlos.

Carlos: Are you Ok?

Ana: Well, I have to write a paper about programming for tomorrow, but I don't understand very well the concept.

Carlos: Why? Can I help you? I am a systems engineer.

Ana: I'm reading my textbook about programming, but the author doesn't define clearly what programming is.

Carlos: There are many postures related to the concept of programming, but the best definition is that programming is the action of lengthening, modifying, and generating the functionality of a system. Both specialists and non-specialists are able to program.

Ana: That's where I get confused. Are you saying that everybody can program?

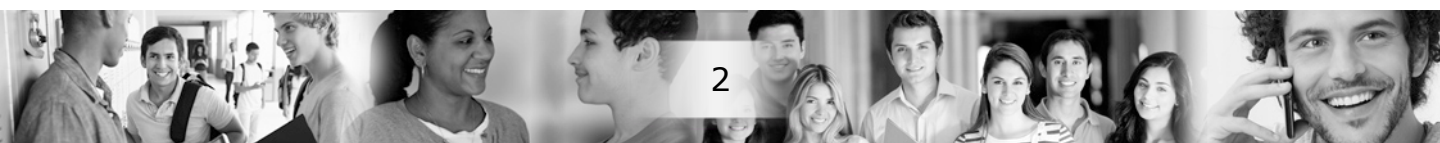
Carlos: Yes, for example, your smartphone; the act of setting a language for your smartphone involves programming; even the act of activating my alarm clock is programming a device. This is broadly the definition of programming.

Ana: I understand, but I want to know, more technically, what programming is.

Carlos: For specialized people, programming refers to the setting steps for a program to run and fulfill certain specifications; in other words, programing aims to design the architecture of a program, and codify it in a suitable programming language. Programming involves both a scientific part and a technical part; the architectural design of a program reflects its scientific conceptualization; while the codification of a program's design represents its technical definitions.

Ana: So, programming is not all about learning the most common programing languages.

Carlos: Exactly! Depending on your design, you use a particular kind of language; for example, social-media application programmers use more complex languages, thus to ease the use of the application; on the other hand, system programmers don't interfere directly with end users, so they have more knowledge of core languages and technical aspects.





Let's Practice!

Ana explains to Jake what computer programming is.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

remember / are able to / have to

1. Computer programming is a general human activity that both experts and common people _____ perform.
2. Pascal is a high level language that application programmers _____ use to ease the handling of the application
3. You need to _____ that computer programming has two essential parts, the scientific and the technical part.



Let's Practice!

Discovering the Concept of Programming

Instructions: Organize the mystery sentences.

1. day. / becomes / every / Computer / programming / simpler

2. applications. / write / languages / programmers / Programming / allow / to

3. HTML. / programming / favorite / language / My / is





Extension Activity

Make your own concept map about computer programming!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for extra information about computer programming.
- Create a concept map explaining what computer programming is, and its characteristics.
- Write sentences about programming using the simple present, the verbs with infinitives and the possessive adjectives.
- Share your concept map with the rest of the class.

Self-Assessment

- I research about computer programming and its characteristics.
- I create a concept map explaining what computer programming is.
- I write sentences in simple present about programming.
- I share my ideas and discuss them with classmates.

Yes

No

☐
☐
☐
☐


Target Words

- **social-media** (noun): application and virtual tools that permit the interaction with other people on internet.
e.g. For example, social-media application programmers use more complex languages, thus to ease the use of the application.
- **broadly** (adverb): in this case, in general words
e.g. This is broadly the definition of programming.
- **lengthen** (verb): the action of expanding, making larger
e.g. There are many postures related to the concept of programming, but the best definition is that programming is the action of lengthening, modifying, and generating the functionality of system.

- **Technical** (adjective): related to the mechanical or industrial crafts.
e.g. Programming involves both a scientific part and a technical part.
- **non-specialists** (noun): a person who is not professional or knowledgeable about a specific field.
e.g. Both specialists and non-specialists are able to program.
- **knowledge** (noun): all what you know.
e.g. System programmers don't interfere directly with end users, so they have more knowledge of core languages and technical aspects.



Learning Points

Verbs + infinitive: Need to and Have to in Affirmative

You can use a verb in infinitive after some verbs to express the purpose of an action.

There are verbs that are followed by another action in the infinitive form to express purpose.

For example:

I need to send some information.

We have to buy a new program.

She wants to sell her computer.

Notice that we conjugate the first verb depending on the subject, but the second verb, the one that expresses the purpose, maintains its infinitive form.

Examples

- *I have to write a paper about programming for tomorrow.*
- *I understand, but I want to know more technically what programming is.*
- *Both specialists and non-specialists are able to program.*

Possessive Adjectives: My - Your - Our in Simple Present Tense

You can use the possessive adjectives to express that something belongs to someone. Possessive adjectives specify the owner.

here are eight possessive adjectives in English, one for each subject pronoun. They are called adjectives because they modify nouns, so you always place them before nouns like things, animals, and people.

For example:

My computers

Your program

Our classroom

Remember that you can use possessive adjectives with the simple present.

Example:

I like your algorithm! It's amazing!

My mother is a famous software programmer.

Her new job at IBM is very interesting

Look at the chart of possessive adjectives.





Learning Points

SUBJECT PRONOUN	POSSESSIVE ADJECTIVES
I	My
You	Your
He	His
She	Her
It	Its
We	Our
You	Your
They	Their

Examples

- *I'm reading my textbook about programming, but the author doesn't define clearly what programming is.*
- *The act of setting a language for your smartphone involves programming.*
- *Depending on your design, you use a particular kind of language.*

Types of Computer Programmers

There are different types of programmers in the systems field. Those software careers vary in the programming languages they use, the purposes they aim, and the relationships they maintain with other colleagues.

Application Programmers

Application programmers write programs for common people to use. They have to use high-level languages that are closer to the English language than to the binary computers' "understanding".

System Programmers

System programmers write programs and codes for operating system software; for that reason, they have less contact with end users than application programmers. As system programmers use core languages closer to the binary code, they have to be more knowledgeable about technical matters.

